How-To Library

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Configure single sign-on to Amazon S3 in Informatica Cloud Data Integration

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Abstract

You can configure single sign-on (SSO) to Amazon S3 with corporate credentials in Cloud Data Integration. SSO provides you with a single point of authentication and an enhanced security to access an Amazon Web Services (AWS) environment.

Supported Versions

- Informatica® Cloud Data Integration Amazon S3 Connector
- Informatica® Cloud Data Integration

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Overview

You can access Amazon S3 with federated single sign-on (SSO) to enable users to authenticate using corporate credentials. SSO provides a mechanism by which users are authenticated once and are then authorized to access other applications. Users are authenticated from a centralized authentication service such as Active Directory (AD).

Advantages of single sign-on

You often need to work with multiple applications for business reasons which are usually provided and hosted by different organizations. In addition, enterprise organizations are increasingly using cloud and third-party SaaS applications. This requires you to remember multiple credentials to access each application that you work with, and this exposes security vulnerabilities and leads to a complex user experience.

Organizations need to control who can access each application without exposing the data security risks. In addition to managing complex user experience and security issues, managing multiple credentials for users in different applications poses a challenge for the IT teams.

Single sign-on offers the following advantages:

Single point of authentication

With federated SSO, the user authentication logic is separated from the application. A trusted Identity Provider (IdP) handles the authentication. The user logs on to the internal web console provided by the identity provider. After a successful authentication, the user gains a token to access other trusted services. This prevents the administrator from working with multiple applications to manage several authentication databases.

Access using corporate credentials

You do not need to create and maintain identities for users in AWS. Amazon S3 can be accessed by using corporate credentials. This avoids maintaining a profile with multiple credentials for different applications.

Enhanced security

Amazon S3 can be accessed by using temporary session credentials. Amazon S3 Connector requests for temporary credentials from AWS STS. The process of fetching temporary credentials happens transparently.

Single sign-on support for Amazon Web Services

SSO is a convenient and safe way to provide access to an Amazon Web Services (AWS) environment. AWS supports federated SSO with Security Assertion Markup Language 2.0 (SAML).

SAML is an open standard for exchanging identity and security information between applications and identity providers like Active Directory Federation Services (ADFS). SAML enables you to sign in using your corporate directory credentials, such as your user name and password, from the Active Directory.

You can log in to the AWS Management Console or invoke the AWS APIs programmatically without requiring the organization to create an Identity and Access Management (IAM) user in AWS.

Transparently, your request is forwarded to the organization's IdP with corporate credentials. Upon successful authentication, the IdP sends a response with the security token. Using this security token, you can either login to the AWS Management Console or invoke programmatic AWS APIs calls by using assertions from an SAML-compliant IdP. The authentication happens without providing any AWS credentials.

Federated single sign-on support in Amazon S3 Connector

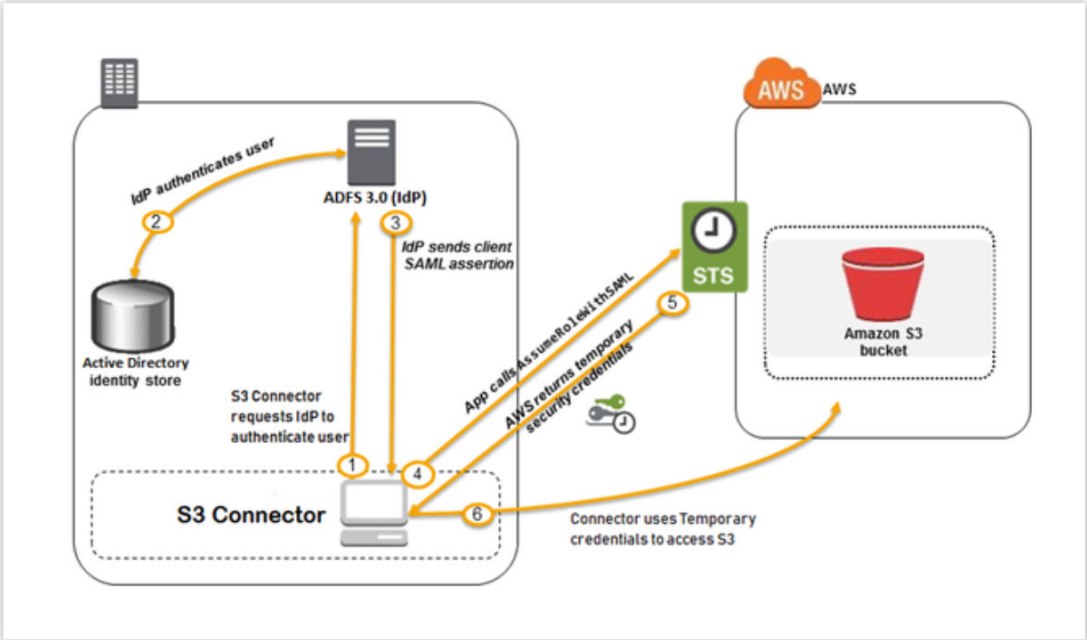
For SAML 2.0-based federation SSO to Amazon S3, configure the first ADFS instance and AWS account to trust each other.

Amazon S3 Connector can access the AWS S3 resources that use temporary security credentials generated for a federated user using SAML 2.0-based SSO authentication.

For information about how to configure ADFS and AWS, see
Enabling Federation to AWS Using Windows Active Directory, ADFS, and SAML 2.0.

Authentication Process

When you configure Amazon S3 Connector for federated SSO, AWS uses the following authentication process:

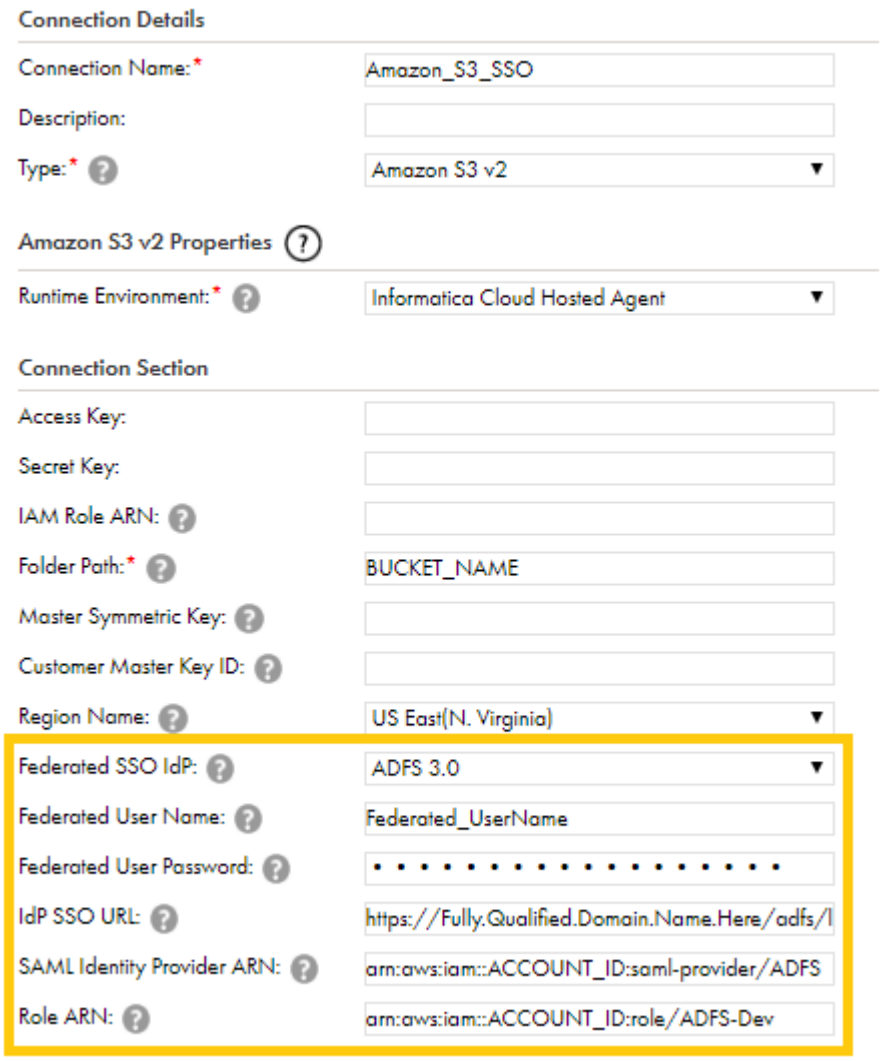


1. Amazon S3 Connector requests authentication for a federated user from the organization's IdP (ADFS 3.0).
2. ADFS 3.0 authenticates the user against the organization's active directory.
3. ADFS 3.0 constructs a SAML assertion with information about the user and sends the assertion to the Amazon S3 Connector as a response to the authentication request.
4. Amazon S3 Connector calls the AWS Security Token Service (STS) `AssumeRoleWithSAML` API. The AWS STS API passes the Amazon Resource Name (ARN) of the SAML provider, the ARN of the role to assume, and the SAML assertion.
5. The AWS STS API response to Amazon S3 Connector includes temporary security credentials.
6. Amazon S3 Connector uses the temporary security credentials to call Amazon S3 API operations.

Connection properties

To use federated SSO, you need to configure certain properties in the Amazon S3 connection.

The following image shows the connection properties that you configure in Amazon S3 Connector for federated SSO:



The following table describes the properties that you need to specify in the Amazon S3 connection:

Property	Description
Folder Path	The bucket name or complete folder path to the Amazon S3 objects.
Region Name	The AWS region in which the bucket is present.
Federated SSO IdP	The SAML 2.0-enabled identity provider for the federated SSO user in AWS.
Federated User Name	The user name for the federated user.
Federated User Password	The password for the federated user.
IdP SSO URL	URL of the organization's identity provider.

Property	Description
SAML Identity Provider ARN	ARN of the SAML identity provider that the AWS administrator created to register the identity provider as a trusted provider.
Role ARN	ARN of the IAM role assumed by the federated user.

For more information about creating an Amazon S3 connection, see Amazon S3 V2 connection properties.

Note: Amazon S3 Connector supports federated SSO for Amazon S3 with ADFS as IDP. Form-based authentication like ADFS version 3.0 and higher are supported.

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